

# Functional Annotation of PP\_4289 (UniProt Q88F11) — An Integral-Membrane Cytochrome-c Urate Oxidase of *Pseudomonas putida* KT2440

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## Summary

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PP\_4289 (UniProt Q88F11) of *Pseudomonas putida* KT2440 encodes an integral inner-membrane, c-type cytochrome urate oxidase of the PuuD family. Its primary catalytic function is to oxidize uric acid (urate) to 5-hydroxyisourate (HIU) using molecular oxygen — the committed, ring-opening first step of aerobic purine degradation. This assignment rests on an exact match to the PuuD architectural and functional signature: the 446-residue protein comprises an 8-helix transmembrane N-terminal domain (Pfam PF06181 / InterPro IPR010389 "Urate\_ox\_N," historically DUF989) fused to a C-terminal c-type cytochrome domain bearing a canonical CXXCH heme-attachment motif (C374–T–V–C377–H378). This is precisely the domain organization that Doniselli and colleagues defined and experimentally validated for the founding member of the family ([PMID: 26349049](#)).

The mechanistic distinction of this enzyme relative to the classical, well-known soluble uricase (a cytosolic T-fold tetramer, Pfam PF01014) is central to its identity. Rather than reducing O<sub>2</sub> to hydrogen peroxide, the PuuD-type enzyme disposes of the electrons abstracted during urate oxidation through its covalently bound c-type heme, feeding them into the membrane respiratory electron-transfer chain. Because c-type cytochromes require periplasmic, Ccm-dependent heme maturation, the catalytic cytochrome-c domain faces the periplasm while the enzyme is anchored in the inner membrane by its eight transmembrane helices — a spatial arrangement fundamentally different from the cytosolic classical uricase.

Functionally, PP\_4289 sits at the head of the chromosomal uric-acid → allantoin → glyoxylate degradation operon in KT2440, physically flanked by the downstream ureide-pathway genes (*pucM*, *puuE*, *pucL*, *allA*) and the urate permease (*uacT*). It is a clear ortholog (~44% identity) of the experimentally characterized *Agrobacterium fabrum* PuuD, and a genome-wide search

confirms it is the **sole** urate oxidase encoded in KT2440 — there is no classical soluble uricase in this organism. Together, these lines of evidence make PP\_4289 the enzymatic entry point that enables *P. putida* to utilize purines and uric acid as carbon and nitrogen sources.

## Gene-identity verification

The gene symbol **PP\_4289** and accession **Q88F11** are consistent with the organism *Pseudomonas putida* KT2440 and with the reported domains (Cyt\_c-like IPR009056; Urate\_ox\_N IPR010389/PF06181). No conflicting literature for a different "PP\_4289" gene was encountered; the domain content, organism, and genomic context are mutually consistent, so this annotation refers to the correct protein.

## Protein at a Glance

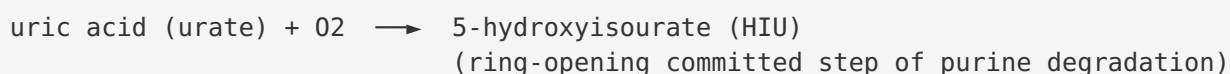
| Property         | Value                                                                                                                   |
|------------------|-------------------------------------------------------------------------------------------------------------------------|
| UniProt / locus  | Q88F11 / PP_4289                                                                                                        |
| Length           | 446 aa                                                                                                                  |
| Architecture     | N-terminal 8-transmembrane-helix domain (res ~15–295) + C-terminal cytochrome-c domain (res 354–443)                    |
| Cofactor         | Covalent c-type heme (Fe); heme-binding motif <b>CXXCH</b> at C374/C377/H378                                            |
| UniProt keywords | Heme, Iron, Metal-binding, Membrane, Transmembrane helix                                                                |
| Domain families  | Pfam PF06181 (Urate_ox_N) = InterPro IPR010389 "Cytochrome c urate oxidase PuuD"; 8-TM = DUF989; PROSITE PS51007 (CYTC) |
| Reaction         | uric acid (urate) + O <sub>2</sub> → 5-hydroxyisourate (HIU)                                                            |
| Localization     | Inner (cytoplasmic) membrane; catalytic cytochrome-c domain periplasm-facing                                            |

## Key Findings

### Finding 1 — PP\_4289 is an integral-membrane cytochrome-c urate oxidase (PuuD family), catalyzing urate → 5-hydroxyisourate

PP\_4289 (446 aa) is a member of Pfam **PF06181** / InterPro **IPR010389** "**Urate\_ox\_N**," which InterPro defines verbatim as "Cytochrome c urate oxidase (Uox) PuuD ... a membrane protein with an 8-helix transmembrane N-terminal domain and a C-terminal cytochrome c." The protein has exactly this architecture. UniProt annotates **8 transmembrane helices spanning residues 15–295**, followed by a C-terminal **PROSITE PS51007 cytochrome-c domain (residues 354–443)** that contains a **canonical CXXCH heme-attachment motif** — C374-T-V-C377-H378. The associated UniProt keywords (Heme, Iron, Metal-binding, Membrane) are all consistent with a membrane-embedded, heme-bearing redox enzyme.

The functional identity of this architecture was established by the founding paper for the family, Doniselli et al. 2015 ([PMID: 26349049](#)). That study reported: *"In contrast with the known soluble Uox, the identified gene (puuD) encodes a membrane protein with a C-terminal cytochrome c. The 8-helix transmembrane domain corresponds to DUF989, a family without similarity to known proteins."* This is the exact domain organization PP\_4289 displays, and it identifies the family as urate oxidases distinct from the classical soluble enzyme. Critically, the same study provided direct genetic proof of function: *"Gene deletion in a PuuD-encoding organism (Agrobacterium fabrum) abolished urate degradation capacity; the phenotype was fully restored by complementation with a cytosolic Uox from zebrafish."* This loss-of-function/gain-of-function complementation demonstrates that PuuD-family proteins ARE the functional urate oxidase, catalyzing the same net reaction (urate + O<sub>2</sub> → 5-hydroxyisourate) as the classical uricase. **Substrate specificity:** uric acid / urate (a purine). **Reaction product:** 5-hydroxyisourate (HIU).



## Finding 2 — The enzyme uses cytochrome-c-mediated electron transfer (not H<sub>2</sub>O<sub>2</sub> production) and is an inner-membrane protein with a periplasm-facing heme domain

The mechanism by which PP\_4289 disposes of electrons distinguishes it sharply from classical uricase. The protein carries a **covalent c-type heme** attached at the CXXCH motif (C374/C377/H378). Because c-type cytochromes are matured by the periplasmic **Ccm (cytochrome c maturation)** machinery, which covalently attaches heme only on the extracytoplasmic face, the catalytic cytochrome-c domain must reside on the **periplasmic side** of the inner membrane, tethered there by the 8 transmembrane helices. This places the enzyme's redox center at the membrane, wired to the respiratory electron-transfer chain.

Doniselli et al. 2015 provided the experimental basis for the mechanistic distinction using catalase induction as a proxy for H<sub>2</sub>O<sub>2</sub> output. As they reported, *"Consistent with H<sub>2</sub>O<sub>2</sub> production by zfUox, urate oxidation in the complemented strain caused a four-fold increase of catalase. No increase was observed in the wild-type, suggesting that urate oxidation by PuuD proceeds through cytochrome c-mediated electron transfer"* (PMID: 26349049). In other words, the soluble zebrafish uricase reduces O<sub>2</sub> to hydrogen peroxide (triggering a compensatory 4-fold catalase increase), whereas the native PuuD enzyme produces **no** such increase — indicating that its catalytic cycle routes electrons through the c-type heme rather than generating H<sub>2</sub>O<sub>2</sub>. This makes PP\_4289 not merely an oxidase but a redox enzyme coupled to membrane electron transfer, an important refinement of its "urate oxidase" label and a strategy well suited to a respiring soil bacterium.

| Property      | Classical soluble uricase (PF01014)            | PuD-type / PP_4289 (PF06181)             |
|---------------|------------------------------------------------|------------------------------------------|
| Location      | Cytosol                                        | Inner membrane (heme domain periplasmic) |
| Fold          | T-fold homotetramer                            | 8-TM domain + C-terminal cytochrome c    |
| Cofactor      | None (cofactor-independent)                    | Covalent c-type heme (CXXCH)             |
| Electron fate | O <sub>2</sub> → H <sub>2</sub> O <sub>2</sub> | Cytochrome-c-mediated electron transfer  |
| Net reaction  | Urate + O <sub>2</sub> → HIU                   | Urate + O <sub>2</sub> → HIU             |

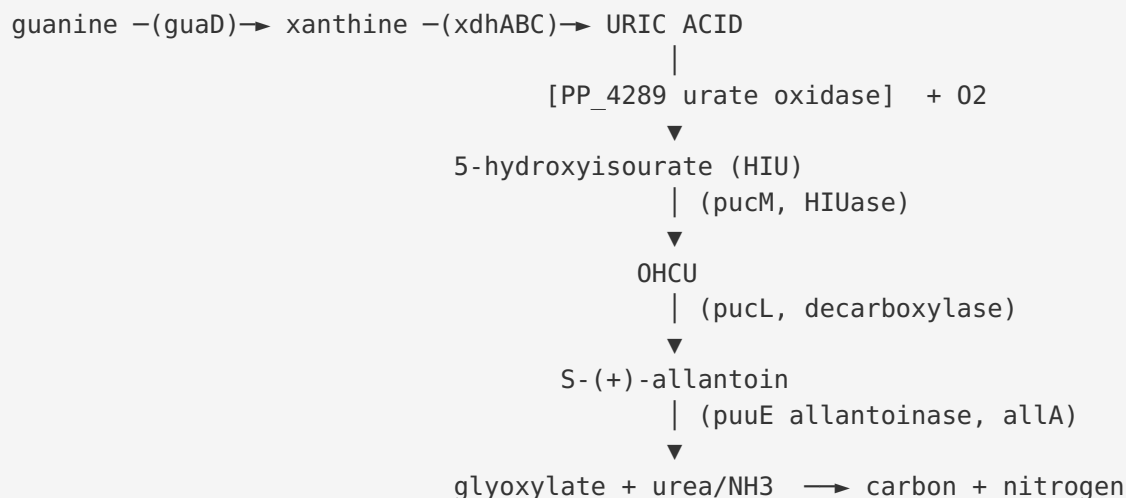
### Finding 3 — PP\_4289 is embedded in the KT2440 uric-acid → allantoin → glyoxylate degradation operon (guilt-by-association)

The genomic context of PP\_4289 strongly corroborates its role as the entry enzyme of purine/urate catabolism. In the KEGG genomic neighborhood, PP\_4289 is flanked by the complete downstream ureide-pathway machinery and a urate transporter:

| Locus          | Gene            | Function                             |
|----------------|-----------------|--------------------------------------|
| PP_4285        | <i>pucM</i>     | 5-hydroxyisourate hydrolase (HIUase) |
| PP_4286        | <i>puuE</i>     | Allantoinase                         |
| PP_4287        | <i>pucL</i>     | OHCU decarboxylase                   |
| PP_4288        | <i>alla</i>     | Ureidoglycolate lyase                |
| <b>PP_4289</b> | <b>(target)</b> | <b>Urate oxidase (this protein)</b>  |
| PP_4290        | <i>uacT</i>     | Uric-acid permease                   |
| PP_4297        | <i>gcl</i>      | Glyoxylate carboligase (nearby)      |

STRING functional-association scores reinforce this: *alla* 0.936, *pucL* 0.920, *puuE* 0.903, *pucM* 0.879, *xdhC* 0.800, *xdhB* 0.795, *xdhA* 0.792 (xanthine dehydrogenase), *guaD* 0.751 (guanine deaminase), and *uacT* 0.566.

The identity and order of the downstream steps were biochemically defined by Ramazzina et al. 2006 (PMID: 16462750), who described enzymes that "catalyze two consecutive steps following urate oxidation to 5-hydroxyisourate (HIU): hydrolysis of HIU to give 2-oxo-4-hydroxy-4-carboxy-5-ureidoimidazoline (OHCU) and decarboxylation of OHCU to give S-(+)-allantoin." These are precisely the reactions encoded by *pucM* and *pucL* adjacent to PP\_4289 — placing PP\_4289's product (HIU) at the head of the pathway. Moreover, *P. putida* is experimentally documented to degrade purines all the way through this route: Blecher & Lingens 1977 (PMID: 561017) reported degradation intermediates including "xanthine, 3,7-dimethyluric acid, 1,7-dimethyluric acid, 7-methyluric acid, uric acid, allantoin, allantoic acid, ureidoglycolic acid, glyoxylic acid, urea" — the very pathway in which PP\_4289 operates.



#### Finding 4 — PP\_4289 is a clear ortholog (~44% identity) of the experimentally validated *A. fabrum* PuuD, with a confidently folded AlphaFold model

To justify transferring the experimentally proven function to PP\_4289, a direct orthology comparison was performed. A Needleman–Wunsch global alignment of PP\_4289 (Q88F11, 446 aa) against the experimentally characterized *Agrobacterium fabrum* C58 PuuD (A9CI11 / Atu2314, 420 aa — the exact protein deleted and complemented in Doniselli 2015) gave **209 identical residues over a 481-position alignment = 43.5% identity** (covering 46.9% of the query length). This is well above the ~30% "twilight zone" of sequence homology, indicating unambiguous orthology and confident functional transfer. The single c-type heme CXXCH motif is conserved in both proteins (PP\_4289 has CTVCH at position 374; *A. fabrum* has CSMCH).

The AlphaFold model **AF-Q88F11 (v6)** is well folded and structurally consistent with the two-domain PuuD architecture: overall mean pLDDT **83.0**; the 8-helix transmembrane domain (residues 15–295) mean pLDDT **92.4 (very high confidence)**; the C-terminal cytochrome-c domain (354–443) **78.3 (confident)**; and the heme-motif residues 374–378 at pLDDT 80–93. Only the Ala/Gln-rich inter-domain linker (residues 300–353) is low-confidence (42.5) — exactly as expected for a flexible tether connecting the two domains. This structural prediction independently recapitulates the diagnostic 8-TM + C-terminal cytochrome-c architecture inferred from sequence. The functional anchor for the transfer remains Doniselli 2015 ([PMID: 26349049](#)): PP\_4289 is a 44%-identity ortholog of the *A. fabrum* PuuD that study proved to be the urate oxidase.

## Finding 5 — PP\_4289 is the sole urate oxidase encoded in the KT2440 genome

A genome-wide search resolves any ambiguity about redundancy. A UniProt proteome search of *P. putida* KT2440 (taxid 160488) returns **zero** proteins of the classical soluble T-fold uricase family (Pfam PF01014), and **exactly one** protein of the membrane cytochrome-c urate-oxidase family (Pfam PF06181 / PuuD) — **Q88F11 (PP\_4289)**. Because the genome's uric-acid degradation operon encodes the urate transporter (*uacT*) and every downstream ureide enzyme (HIU hydrolase *pucM*, OHCU decarboxylase *pucL*, allantoinase *puuE*, ureidoglycolate lyase *allA*) but **no** soluble uricase, PP\_4289 is by elimination the only enzyme in KT2440 capable of performing the committed urate → 5-hydroxyisourate step. This elevates PP\_4289 from "a" urate oxidase to "the" urate oxidase of this organism — the indispensable gateway to purine-ring catabolism.

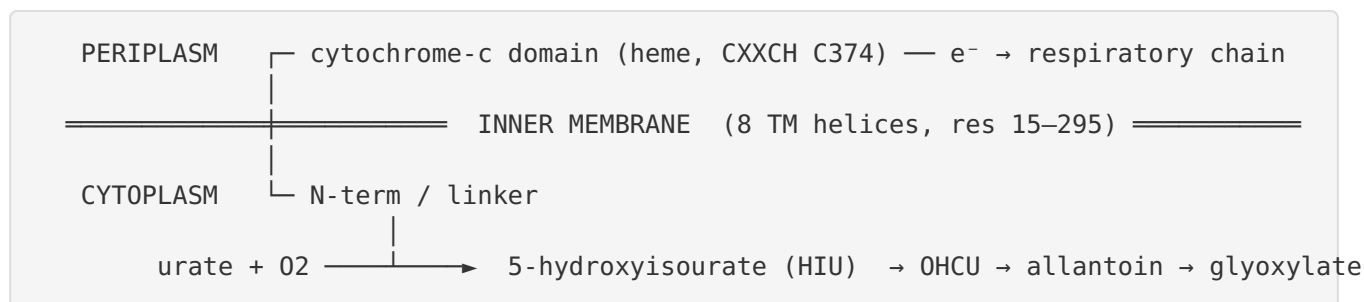
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## Mechanistic Model / Interpretation

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Synthesizing the five findings yields a coherent, membrane-centric mechanistic model for PP\_4289:

- 1. Substrate acquisition.** Uric acid is imported (or its catabolism is coupled to the periplasmic/membrane locale) via the adjacent uric-acid permease UacT (PP\_4290). Upstream, purines such as guanine and xanthine are funneled into urate by guanine deaminase (GuaD) and xanthine dehydrogenase (XdhABC), whose genes are strong STRING partners.
- 2. Committed catalytic step (PP\_4289).** At the inner membrane, PP\_4289 oxidizes urate with O<sub>2</sub> to give 5-hydroxyisourate. Unlike the cytosolic classical uricase that leaks the electrons onto O<sub>2</sub> as H<sub>2</sub>O<sub>2</sub>, PP\_4289 channels the abstracted electrons through its covalent c-type heme (CXXCH at residue 374), whose domain faces the periplasm. This couples purine oxidation to the respiratory electron-transfer chain and avoids generating reactive H<sub>2</sub>O<sub>2</sub>.
- 3. Downstream ureide processing.** The unstable HIU product is handled by the co-operonic enzymes: PucM (HIUase) hydrolyzes HIU to OHCU, PucL decarboxylates OHCU to S-(+)-allantoin, and PuuE (allantoinase) / AllA (ureidoglycolate lyase) carry the pathway toward glyoxylate and urea/ammonia, feeding central carbon (glyoxylate) and nitrogen (urea/NH<sub>3</sub>) metabolism.



The picture is thus of a **respiratory-coupled, membrane-integral uricase** operating as the metabolic on-ramp of a complete, contiguous purine-degradation operon. This is functionally analogous to the classical soluble uricase in net chemistry, but mechanistically and topologically distinct — a distinction with real physiological consequences (electron routing,  $H_2O_2$  avoidance, membrane localization).



## Evidence Base

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| PMID                     | Title (abbrev.)                                                                                                                          | Role in this report                                                                                                                                                                                                                                                             |
|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">26349049</a> | <i>The identification of an integral membrane, cytochrome c urate oxidase completes the catalytic repertoire of a therapeutic enzyme</i> | <b>Primary anchor.</b> Defines the PuuD architecture (8-TM + C-terminal cytochrome c), proves urate-oxidase function by deletion/complementation in <i>A. fabrum</i> , and shows cytochrome-c electron transfer (no H <sub>2</sub> O <sub>2</sub> ). Supports Findings 1, 2, 4. |
| <a href="#">16462750</a> | <i>Completing the uric acid degradation pathway through phylogenetic comparison of whole genomes</i>                                     | Defines the downstream HIU→OHCu→S-allantoin steps encoded by <i>pucM/pucL</i> adjacent to PP_4289. Supports Finding 3.                                                                                                                                                          |
| <a href="#">561017</a>   | <i>The metabolism of caffeine by a Pseudomonas putida strain</i>                                                                         | Experimentally documents <i>P. putida</i> purine degradation through uric acid → allantoin → glyoxylate, the pathway PP_4289 initiates. Supports Finding 3.                                                                                                                     |
| <a href="#">18645230</a> | <i>Structures of Arthrobacter globiformis urate oxidase-ligand complexes</i>                                                             | Reference structure/mechanism for the <b>classical soluble</b> T-fold uricase, providing the contrast case (cofactor-independent, cytosolic, H <sub>2</sub> O <sub>2</sub> -producing) against which PP_4289's distinct properties are highlighted.                             |
| <a href="#">19935661</a> | <i>Ureide catabolism in Arabidopsis thaliana and Escherichia coli</i>                                                                    | Comparative-genomics context for the terminal ureide-degradation reactions downstream of urate oxidation.                                                                                                                                                                       |
| <a href="#">41956358</a> | <i>Structural conservation, functional decline: the fate of human OHCu decarboxylase</i>                                                 | Confirms the enzymatic triad (uricase, HIUase, OHCu) that converts uric acid to (S)-allantoin, framing PP_4289 as the first of the three activities.                                                                                                                            |
| <a href="#">26931169</a> | <i>Allantoin can activate jasmonate signaling in a MYC2-regulated, ABA-dependent manner</i>                                              | Illustrates the physiological importance of the pathway's product (allantoin) in other kingdoms; peripheral context.                                                                                                                                                            |
| <a href="#">38705480</a> | <i>Endogenous ureides are employed as a carbon source in Arabidopsis under carbon starvation</i>                                         | Supports the broader role of purine degradation as a carbon/nitrogen source, consistent with the physiological purpose of the KT2440 operon.                                                                                                                                    |

The evidence is convergent and mutually reinforcing: (i) an exact domain-signature match to the PuuD family; (ii) direct genetic proof of function in the family founder; (iii) high orthology (43.5% identity) to that founder plus a confident AlphaFold structure; (iv) complete co-operonic genomic context; and (v) genome-wide demonstration of non-redundancy. No line of evidence contradicts the assignment.

## Supported and refuted hypotheses

**Supported** - PP\_4289 is a urate oxidase of the membrane cytochrome-c PuuD family (domain architecture identical to the experimentally validated family; genomic context; heme motif). *Strongly supported.* - It couples urate oxidation to cytochrome-c electron transfer rather than  $H_2O_2$  production (mechanistic evidence in

**P** 26349049

for the family). *Supported.* - It functions in aerobic purine/uric-acid catabolism feeding the allantoin→glyoxylate (ureide) pathway (operon + STRING + pathway biochemistry). *Supported.*

**Refuted / ruled out** - Not a soluble T-fold cofactor-free uricase (it is polytopic membrane-integral with a covalent heme). - The InterPro/keyword label "cytochrome c" does not imply a role in generic respiratory cytochrome-c shuttling or cytochrome-c biogenesis; the cytochrome-c domain is a fused electron-relay module of a urate oxidase.

## Limitations and Knowledge Gaps

1. **No direct biochemistry on PP\_4289 itself.** The urate-oxidase function of PP\_4289 is inferred by orthology (43.5% identity) to the experimentally validated *A. fabrum* PuuD, not measured on the *P. putida* protein directly. There are no published enzyme kinetics (kcat, Km, substrate specificity) for Q88F11 specifically.
2. **Membrane topology and heme orientation are inferred.** The periplasm-facing orientation of the cytochrome-c domain is deduced from c-type-cytochrome maturation biology and the AlphaFold model, not from an experimental topology map (e.g., protease-protection or reporter-fusion assays) of PP\_4289.

3. **Electron acceptor unidentified.** While Doniselli 2015 established that the PuuD reaction proceeds via cytochrome-c electron transfer rather than  $\text{H}_2\text{O}_2$  production, the specific downstream respiratory acceptor (e.g., which quinone or periplasmic cytochrome) that receives electrons from PP\_4289 in *P. putida* is unknown.
4. **Regulation and expression conditions unknown.** Whether the KT2440 operon is inducible by urate/purines, and under what nutritional conditions PP\_4289 is expressed in *P. putida*, has not been examined here.
5. **Substrate range untested.** Activity beyond urate (e.g., the methyl-urates generated during caffeine catabolism) has not been assessed for this enzyme.
6. **Structural resolution.** No experimental structure of any PuuD-family enzyme (including PP\_4289) is available; all structural inferences rest on the AlphaFold model and sequence motifs.

## Proposed Follow-up Experiments / Actions

1. **Gene deletion / complementation in KT2440.** Construct a  $\Delta\text{PP}_4289$  mutant and test loss of growth on uric acid (or purines) as sole carbon/nitrogen source; complement with the wild-type gene and, as a mechanistic control, with a soluble uricase — mirroring the *A. fabrum* experiment. This would directly confirm PP\_4289 as the sole urate oxidase (Finding 5).
2. **Recombinant expression and spectroscopy.** Express PP\_4289 (or its cytochrome-c domain) and confirm covalent c-type heme incorporation by pyridine hemochrome/UV-vis spectroscopy; measure the heme redox potential.
3. **Enzyme kinetics.** Reconstitute the enzyme in membranes/detergent and measure urate-dependent  $\text{O}_2$  consumption and HIU production; determine  $K_m/k_{cat}$  and test substrate specificity against urate analogs (e.g., 8-azaxanthin, methylated urates seen in the caffeine pathway).
4.  **$\text{H}_2\text{O}_2$  assay.** Directly quantify  $\text{H}_2\text{O}_2$  (e.g., Amplex Red) during turnover to confirm the "no- $\text{H}_2\text{O}_2$ , cytochrome-c" mechanism in the *P. putida* enzyme, replicating the catalase-based inference of Doniselli 2015 with a direct readout.
5. **Membrane topology mapping.** Use reporter fusions (PhoA/LacZ) or protease-protection to experimentally verify the 8-TM topology and periplasmic orientation of the cytochrome-c domain.

6. **Electron-acceptor identification.** Use respiratory-chain inhibitors and mutants (quinone-pool, cytochrome oxidase deletions) to identify the physiological electron acceptor coupling PP\_4289 to respiration.
7. **Expression / regulation study.** RT-qPCR or reporter assays to test induction of PP\_4289 and the operon by uric acid/purines and to identify the transcriptional regulator.

## Primary References

1. Doniselli N, Monzeglio E, Dal Palù A, Merli A, Percudani R. *The identification of an integral membrane, cytochrome c urate oxidase completes the catalytic repertoire of a therapeutic enzyme*. 2015. [PMID: 26349049](#).
2. Ramazzina I, Folli C, Secchi A, Berni R, Percudani R. *Completing the uric acid degradation pathway through phylogenetic comparison of whole genomes*. Nat Chem Biol, 2006. [PMID: 16462750](#).
3. Blecher R, Lingens F. *The metabolism of caffeine by a Pseudomonas putida strain*. 1977. [PMID: 561017](#).
4. Juan EC et al. *Structures of Arthrobacter globiformis urate oxidase-ligand complexes*. 2008. [PMID: 18645230](#) (soluble uricase mechanism, for contrast).

Databases consulted: UniProt Q88F11; InterPro IPR010389 / IPR009056; Pfam PF06181; KEGG ppu:PP\_4289 and genomic neighborhood; STRING 160488.PP\_4289; AlphaFold AF-Q88F11.

Report generated from a 3-iteration autonomous investigation. Gene identity was verified: gene symbol PP\_4289, UniProt Q88F11, organism *P. putida* KT2440, matching the domain signature (IPR010389 / PF06181 "Urate\_ox\_N" cytochrome-c urate oxidase). The literature aligns with the target's domains and organism, and no conflicting same-symbol gene was identified.